

SH1017 Physics for Engineering Mathematics Lab 3

Lucas Frykman

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Q1

Method: We numerically integrate the time-independent Schrödinger equation using the Verlet method (`eigenstates.py`), enforcing boundary conditions via the shooting method. We then evaluate the wave function at the boundary wall, calculating $\Delta = |\psi(1)|$, using the theoretical ground state energy $E = 0.5\pi^2$ for the infinite box potential across varying mesh sizes $N \in [10, 100, 1000, 10000, 100000]$.

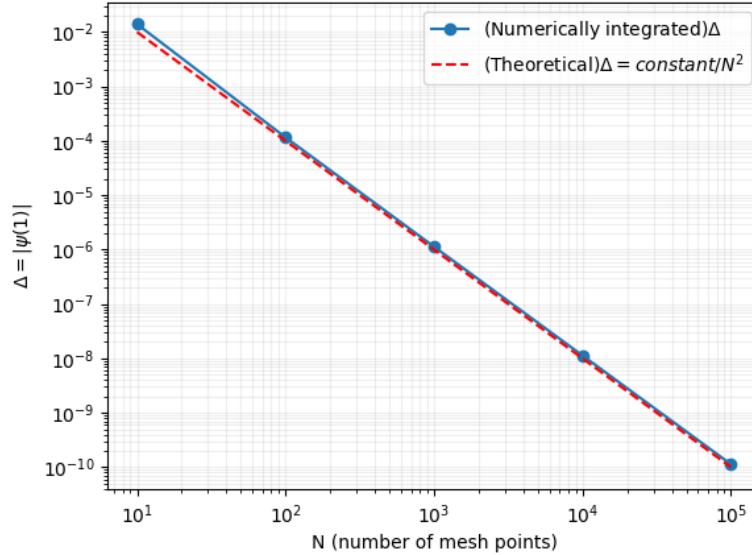


Figure 1: Log-log plot of Δ vs N . Parameters: $E = 0.5\pi^2$, $N \in \{10, 100, 1000, 10000, 100000\}$, $dx = 1/N$.

Theory: The probability of finding the particle at the infinite wall is zero ($\Delta = 0$). However Verlet numerical integration introduces an error that decreases quadratically with the number of mesh points N . Therefore, we expect $\Delta = \text{constant}/N^2$, which corresponds to a linear relationship in a log-log plot with a slope of roughly -2 , since $\log(\Delta) = \log(k) - 2\log(N)$.

Results: The plotted numerical data in the log-log plot shows an approximately linear relationship that is consistent with the theoretical slope of -2 . This supports the expected quadratic convergence of the Verlet numerical integration method.

Q2

Method: We calculate the energy eigenvalues for an infinite box potential (where $V(x) = 0$ inside the box and infinite outside). We use a binary search algorithm, paired with the shooting method, with a tolerance of 10^{-9} and $N = 10000$ mesh points to bracket the 4 lowest energy eigenvalues and plot their respective eigenstates.

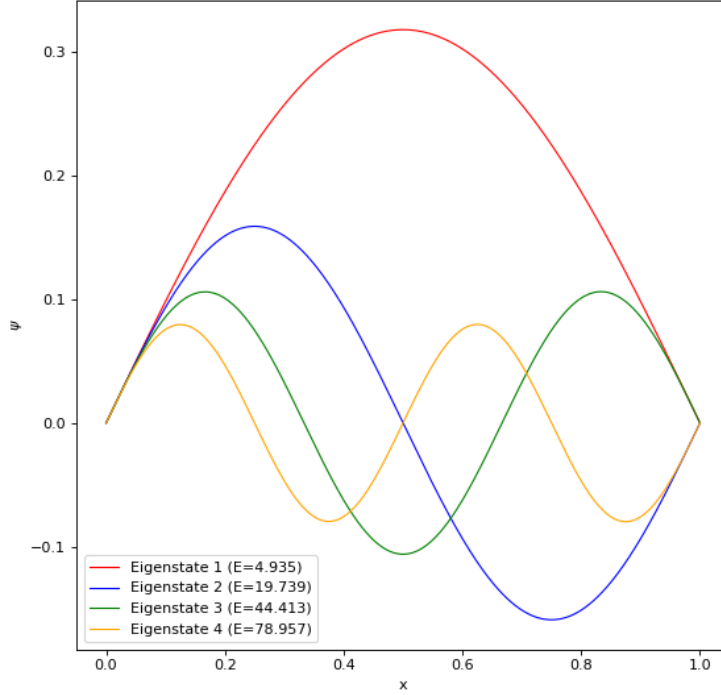


Figure 2: The 4 lowest eigenstates for $V = 0$. Parameters: $N = 10000$, $dx = 1/N$, tolerance = 10^{-9} .

Theory: For an infinite box, the theoretical wavefunctions must be exactly zero at the boundaries ($\psi(1) = 0$). Furthermore, higher energy states possess a larger momentum p . According to the de Broglie relation $\lambda = \frac{h}{p}$, a larger momentum implies a shorter wavelength, leading to more rapid spatial oscillations in the wavefunction.

Results: The calculated lowest energy eigenvalues are $E_1 = 4.935$, $E_2 = 19.739$, $E_3 = 44.413$, and $E_4 = 78.957$. The plotted eigenstates show that $\psi(1) \approx 0$ at the boundary. In agreement with the de Broglie theory, we observe that the number of oscillations in the wavefunction increases consistently with the energy state n .

Q3

Method: We simulate the harmonic oscillator with the potential $V(x) = \frac{1}{2}x^2$ using $N = 10000$ mesh points and a spatial step $dx = 5/N$. We compute the lowest four energy eigenvalues using the binary search and shooting method, taking advantage of the parity of the solutions.

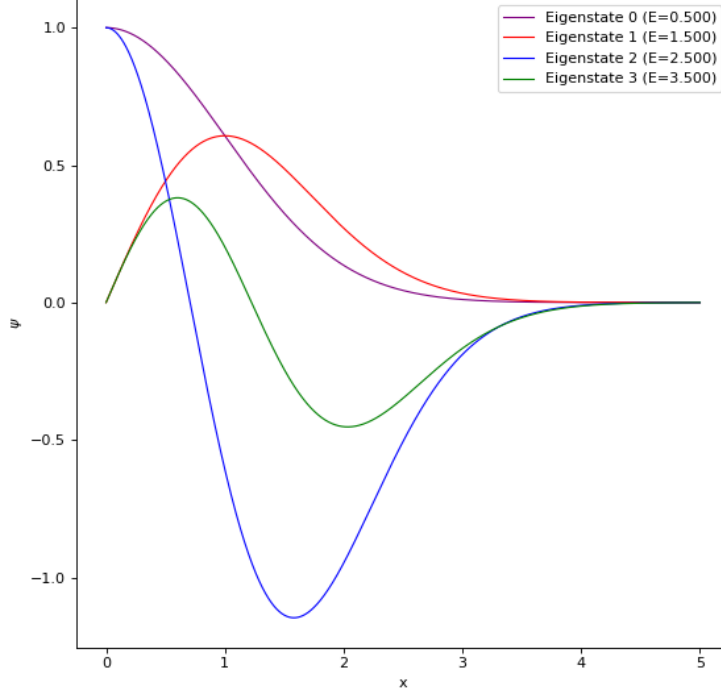


Figure 3: The 4 lowest eigenstates for the harmonic oscillator. Parameters: $N = 10000$, $dx = 5/N$, tolerance $= 10^{-9}$.

Theory: The time independent schrödinger equation for the harmonic oscillator is $-\frac{1}{2} \frac{d^2 \psi}{dx^2} + \frac{1}{2} x^2 \psi = E \psi$. This can be solved analytically using Hermite polynomials $H_n(x)$, yielding the exact solutions $\psi_n(x) = A_n e^{-x^2/2} H_n(x)$ and exact energy eigenvalues $E_n = n + 1/2$. The eigenstates must satisfy parity boundary conditions at $x = 0$: even states (n even) have $\psi(0) = 1, \psi'(0) = 0$, while odd states (n odd) have $\psi(0) = 0, \psi'(0) = 1$. All physical wavefunctions must decay to zero as $x \rightarrow \infty$.

Results: The numerically calculated energy eigenvalues are exactly $E_0 = 0.500, E_1 = 1.500, E_2 = 2.500$, and $E_3 = 3.500$, which is consistent with the theoretical values $E_n = n + 1/2$. Additionally, the plotted eigenstates visibly satisfy the expected parity conditions at the origin ($x = 0$) and correctly decay toward zero at large distances ($x \rightarrow 5$).

Q4

Method: We calculate the 4 lowest energy eigenvalues for an anharmonic potential $V(x) = \frac{1}{2}x^2 + \frac{1}{2}x^4$. We then compute the theoretical first-order perturbation approximation where the unperturbed Hamiltonian is $H_0 = -\frac{1}{2} \frac{d^2}{dx^2} + \frac{1}{2}x^2$ and the perturbation is $H_1 = \frac{1}{2}x^4$.

n	Numerical E	Perturbation E
0	0.696	0.875
1	2.325	3.375
2	4.328	7.375
3	6.579	12.875

Table 1: Energy eigenvalues for the anharmonic potential.

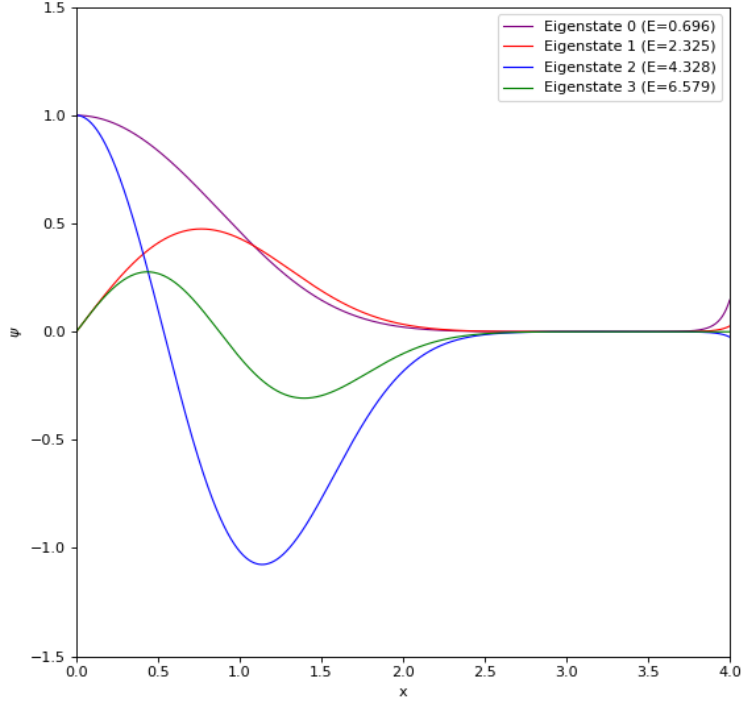


Figure 4: The 4 lowest eigenstates for the anharmonic potential. Parameters: $N = 10000$, $dx = 5/N$, tolerance $= 10^{-9}$.

Theory: First-order perturbation theory provides the energy correction $E_n^{(1)} = \langle \psi_n^{(0)} | H_1 | \psi_n^{(0)} \rangle$, yielding corrected eigenvalues $E_n \approx E_n^{(0)} + E_n^{(1)}$. This approximation is generally only valid and accurate when the perturbation is relatively small compared to the unperturbed Hamiltonian.

Results: In figure 4, the wave functions decay towards zero (similarly to figure 3) but then start to blow up near $x = 4$. The numerical results in the table show that the first-order perturbation approximation consistently overestimates the energy eigenvalues. As n increases, the deviation between the numerical results and the perturbation method grows significantly.